



Universität Hamburg

DER FORSCHUNG | DER LEHRE | DER BILDUNG



Universitätsklinikum
Hamburg-Eppendorf

Predicting Protein Crystallization Conditions

Data Acquisition, Preprocessing, Exploratory Data Analysis

Michael Hüppe



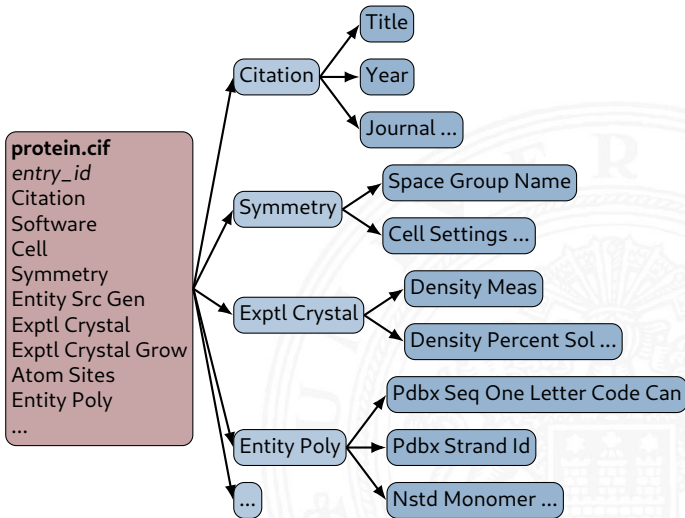
University of Hamburg
Faculty of Mathematics, Informatics and Natural Sciences
Department of Informatics

Maschinelles Lernen in der Bioinformatik

November 28, 2025



1. Filter out entries not crystallized using X-Ray Crystallography 10%
2. Filter out entries which did not have any crystallization information
3. Restructure database from monolithic to relational



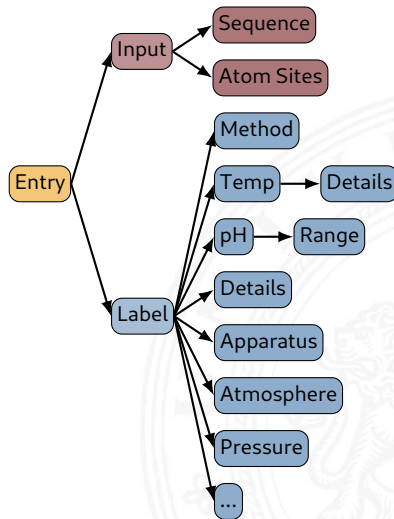
Number of entries: 196743

Number of features: 612

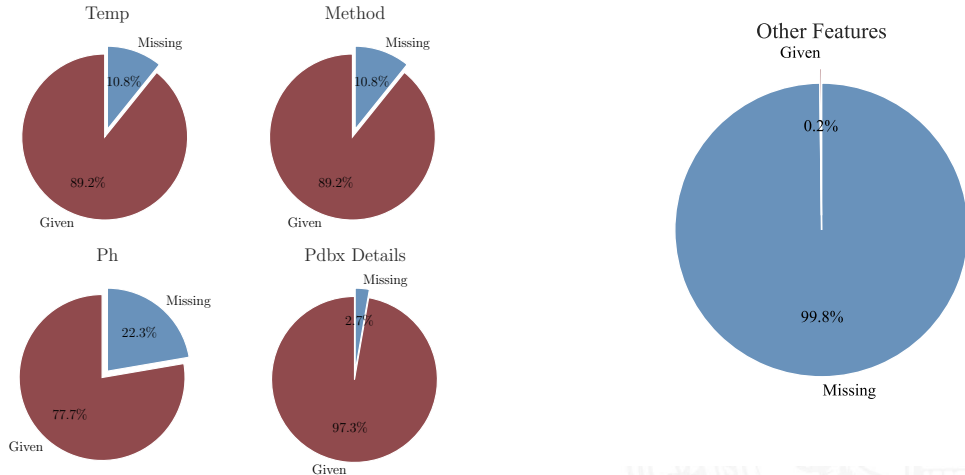
Features related to **input** : 2

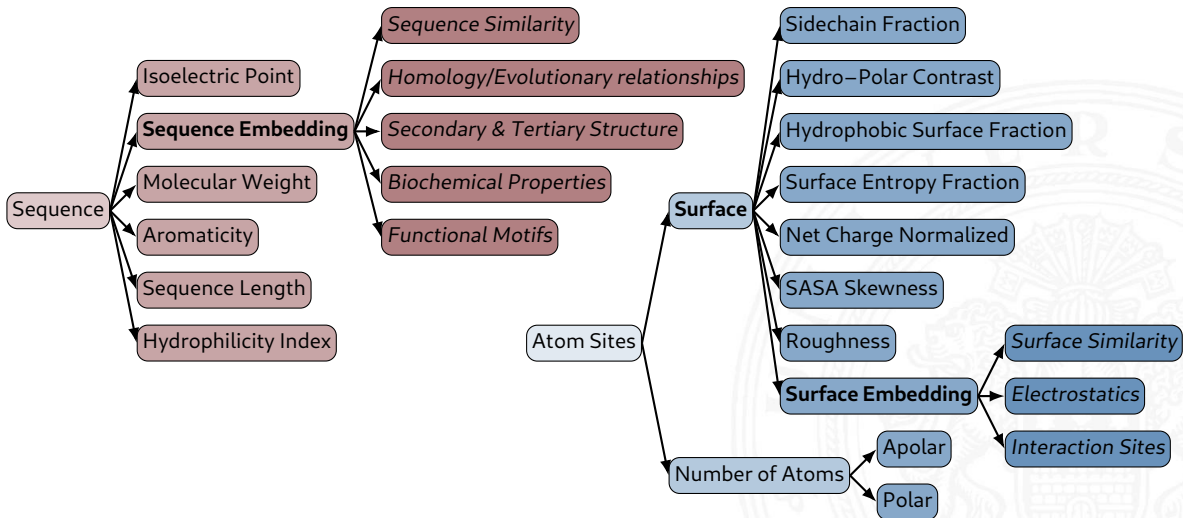
Features related to **label** : 18

BUT!



Label





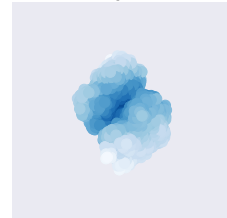
Sequence Embedding

- ▶ ESM based from Rives et al., 2021
- ▶ trained on 250m protein sequences
- ▶ deep Transformer learned using masked language modeling objective
- ▶ biochemical properties of amino acids to remote homology of proteins

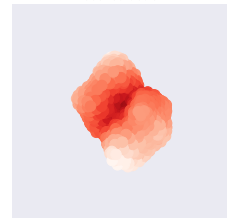
Surface Embedding

- ▶ trained on 200k protein structures
- ▶ surface sampled using dMaSif (Sverrisson et al., 2020)
- ▶ Auto-Encoder approach to create surface embedding

Original



Reconstruction



Sequence Embedding

- ▶ ESM based from Rives et al., 2021
- ▶ trained on 250m protein sequences
- ▶ deep Transformer learned using masked language modeling objective
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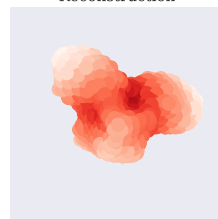
Surface Embedding

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- ▶ surface sampled using dMaSif (Sverrisson et al., 2020)
- ▶ Auto-Encoder approach to create surface embedding

Original



Reconstruction



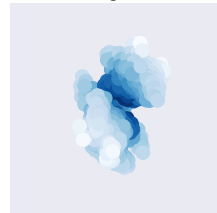
Sequence Embedding

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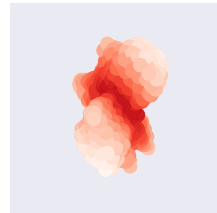
Surface Embedding

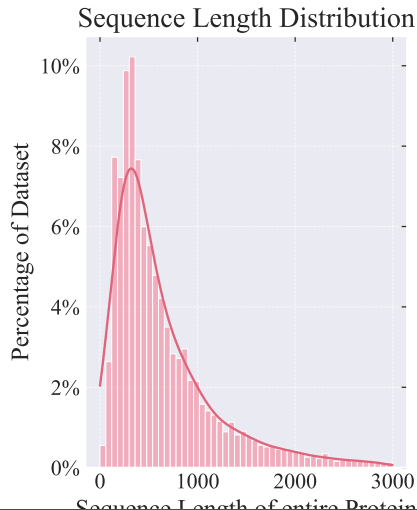
- ▶ trained on 200k protein structures
- ▶ surface sampled using dMaSif (Sverrisson et al., 2020)
- ▶ Auto-Encoder approach to create surface embedding

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Reconstruction

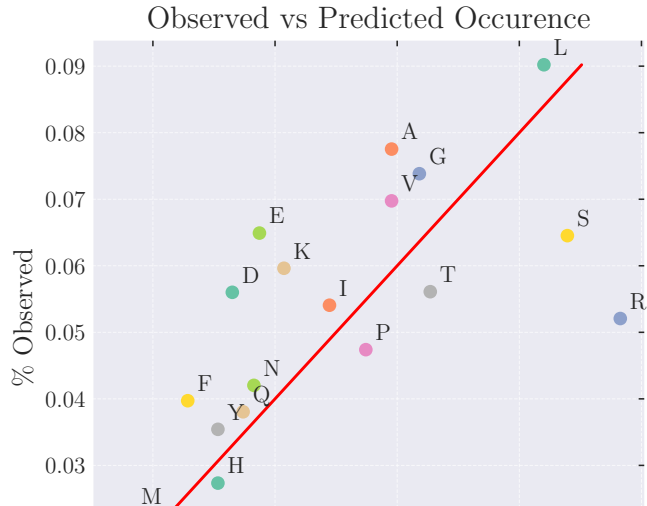
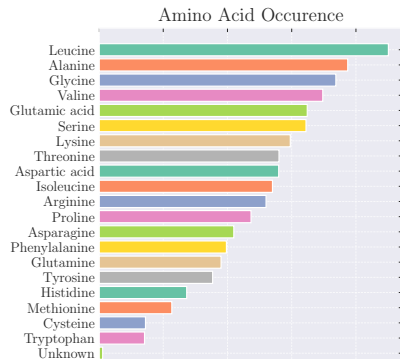


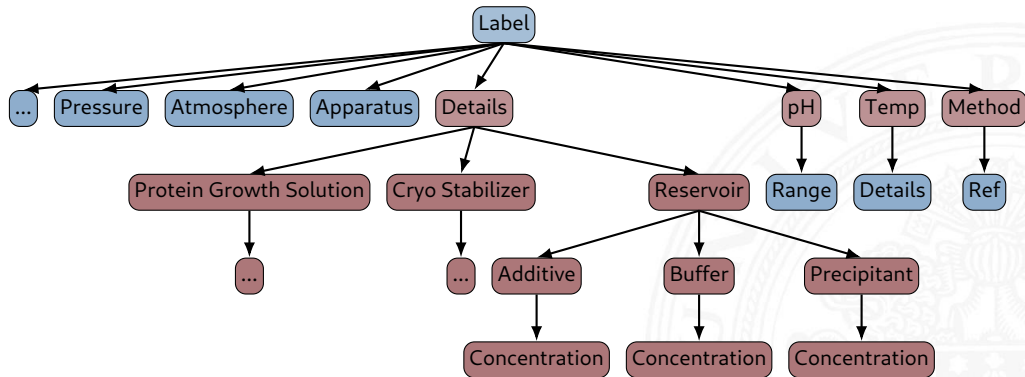


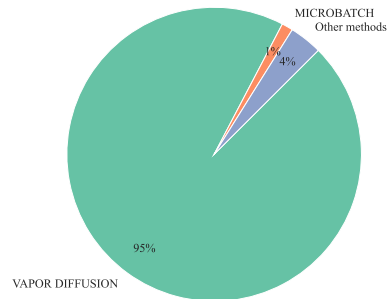
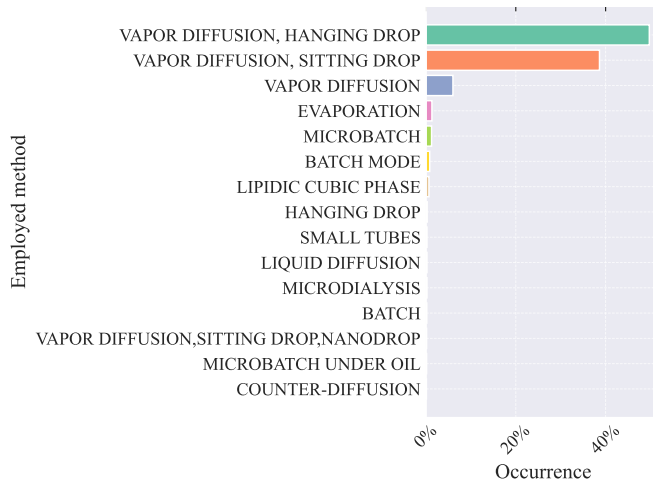
Sequence Length Distribution

- ▶ Input size has to be chosen such that the majority of the protein space is included in the dataset
- ▶ Outliers might unnecessarily increase model size or are too small to encode information (92% for [60, 1000])

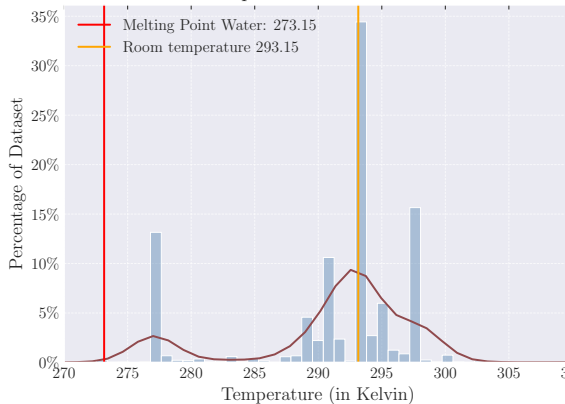
Sequence Length	
count	280543
mean	262
std	191
min	2
max	4128



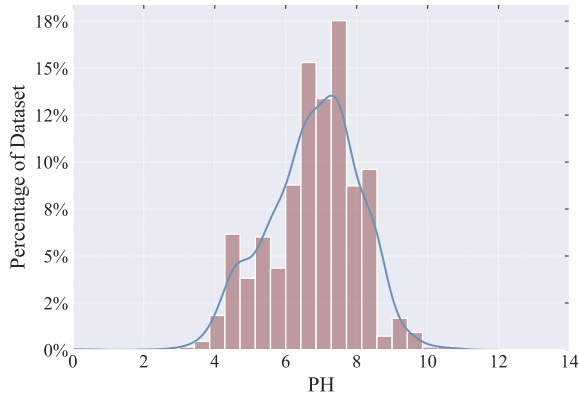




Temperature Distribution



PH Distribution



Based on preprocessing proposed by Lynch, Dudek, and Bowman, 2020

Crystallization was carried out in sitting-drop vapor-diffusion setups with 1:1 mixtures of protein solution containing 0.7 mM hepes and 1.8 mM MnCl₂ and reservoir solution containing 19%-21% PEG 3350 and 30 **percent polyethylene glycol monomethyl ether** 4000 and 0.2 M Na₂HPO₄ , pH 9.5, VAPOR DIFFUSION, SITTING DROP, temperature 298K



Crystallization was carried out in sitting-drop vapor-diffusion setups with 1:1 mixtures of protein solution containing 0.7 mM hepes and 1.8 mM MnCl₂ and reservoir solution containing 19%-21% PEG 3350 and 30% **PEGMME** 4000 and 0.2 M Na₂HPO₄ , pH 9.5, VAPOR DIFFUSION, SITTING DROP, temperature 298K

Crystallization was carried out in sitting-drop vapor-diffusion setups with 1:1 mixtures of **protein solution** containing 0.7 mM heppeess and 1.8 mM MnCl₂ and **reservoir solution** 19%-21% PEG 3350 and 30% PEGMME 4000 and 0.2 M Na₂HPO₄ , **pH 9.5, VAPOR DIFFUSION, SITTING DROP, temperature 298K**



protein solution : containing 0.7 mM hepes and 1.8 mM MnCl₂ and
reservoir solution : containing 19%-21% PEG 3350 and 30% PEGMME 4000 and 0.2 M Na₂HPO₄
pH 9.5 ,
VAPOR DIFFUSION, SITTING DROP ,
temperature 298K

protein solution: containing 0.7 mM hepes and 1.8 mM MnCl₂ and
reservoir solution: containing **19%-21%** PEG 3350 and 30% PEGMME 4000 and
0.2 M Na₂HPO₄ ,
pH 9.5,
VAPOR DIFFUSION, SITTING DROP,
temperature 298K



protein solution: containing 0.7 mM hepes and 1.8 mM MnCl₂ and
reservoir solution: containing **20%** PEG 3350 and 30% PEGMME 4000 and 0.2
M Na₂HPO₄ ,
pH 9.5,
VAPOR DIFFUSION, SITTING DROP,
temperature 298K

protein solution: containing 0.7 mM hepes and 1.8 mM MnCl₂ and
reservoir solution: containing 20% PEG 3350 and 30% PEGMME 4000 and **0.2 M** Na₂HPO₄,
pH 9.5,
VAPOR DIFFUSION,
SITTING DROP, temperature 298K



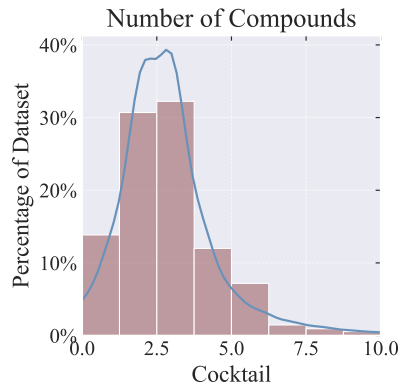
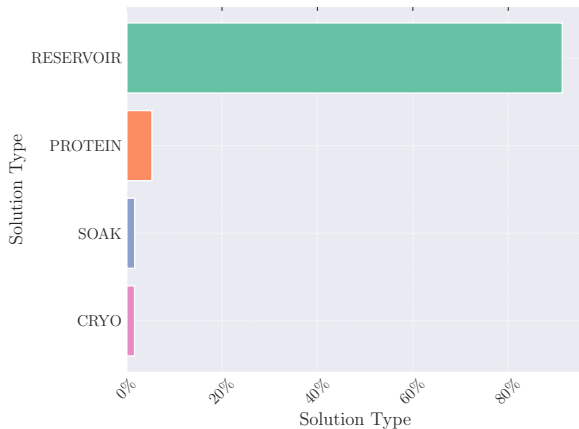
protein solution: containing 0.7 mM hepes and 1.8 mM MnCl₂ and
reservoir solution: containing 20% PEG 3350 and 30% PEGMME 4000 and **200 mM** Na₂HPO₄,
pH 9.5,
VAPOR DIFFUSION, SITTING DROP,
temperature 298K

protein **solution** : **containing** 0.7 mM hepes **and** 1.8 mM MnCl₂ **and**
reservoir **solution** : **containing** 20% PEG 3350 **and** 30% PEGMME 4000 **and** 0.2
M Na₂HPO₄ ,
pH 9.5,
VAPOR DIFFUSION, SITTING DROP,
temperature 298K



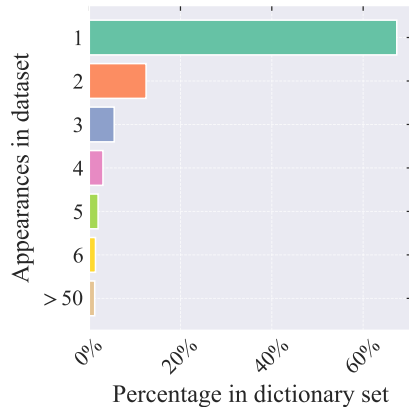
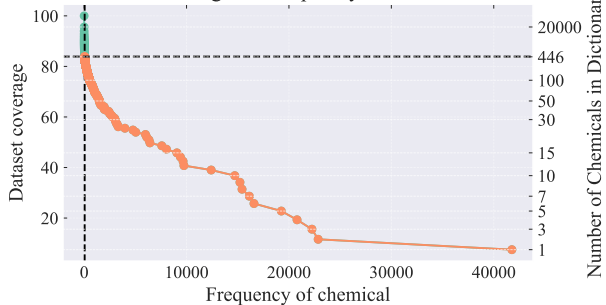
protein: 0.7 mM hepes 1.8 mM MnCl₂
reservoir: 20% PEG 3350 30% PEGMME 4000 200 mM Na₂HPO₄ ,
pH 9.5,
VAPOR DIFFUSION, SITTING DROP,
temperature 298K

```
{  
  "protein": {  
    "hepes": 0.7mM,  
    "MnCl2": 1.8mM,  
  
    },  
  "reservoir": {  
    "PEG 3350": 20%,  
    "PEGMME 4000": 30%,  
    "Na2HPO4": 200mM,  
  
    },  
  "pH": 9.5,  
  "method": Vapor Diffusion, Sitting Drop  
  "temperature": 298k  
}
```

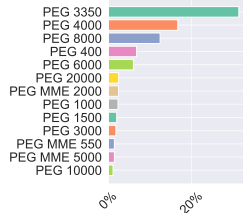


- ▶ Number of unique chemicals: 24999
- ▶ After frequency filtering at 100: 348

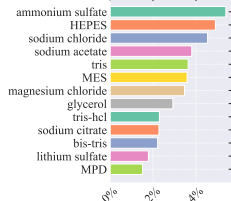
Coverage vs. frequency threshold



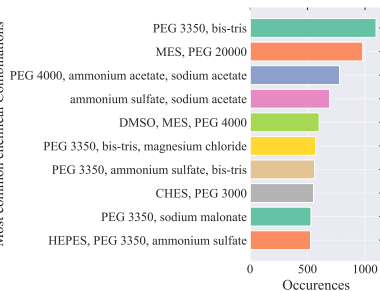
Most Common PEG



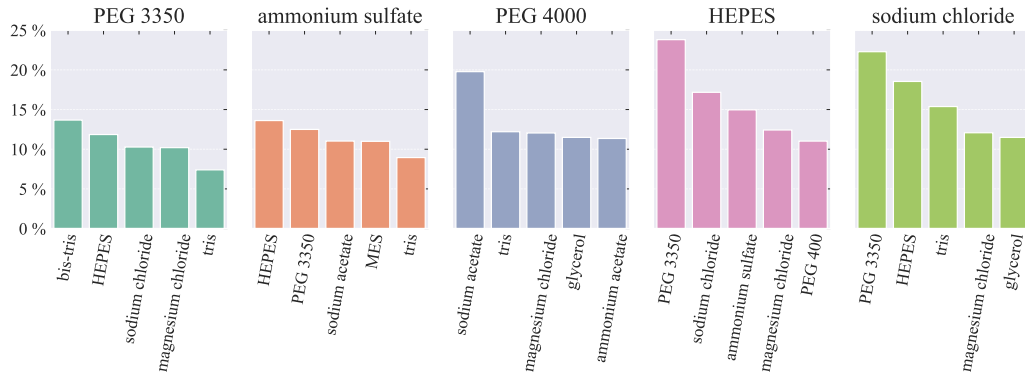
Most Common Chemical



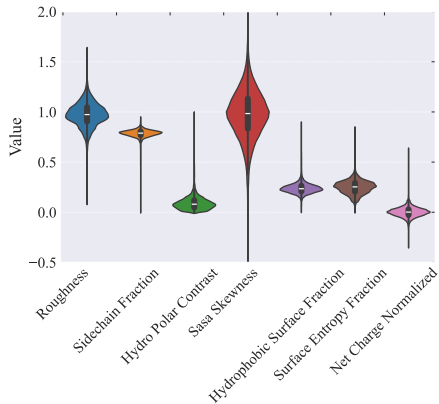
Most common chemical combinations



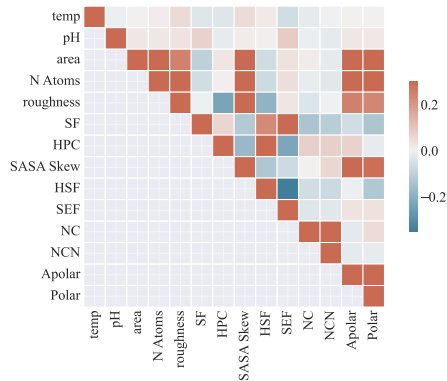
Most Common 5-compound subsets within chemical sets

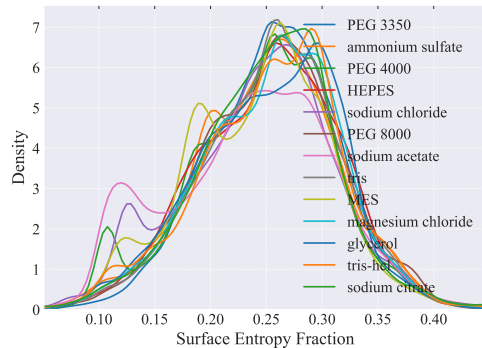
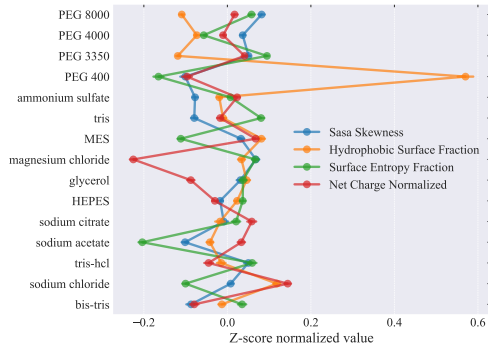


Distribution of Surface Information



Correlation Matrix between surface information and pH/temp





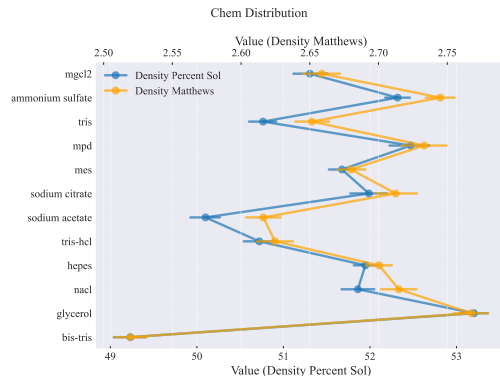
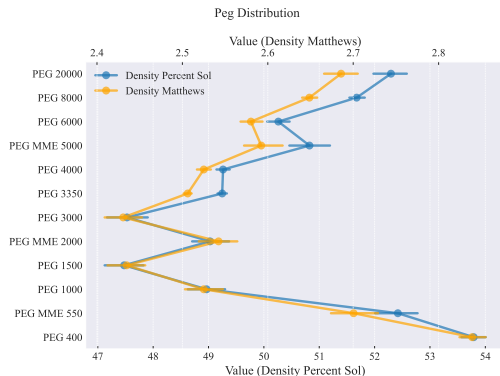
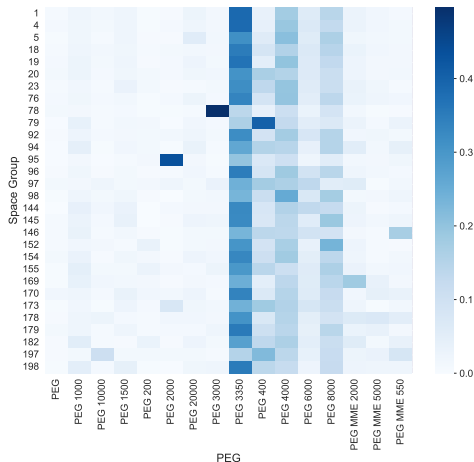
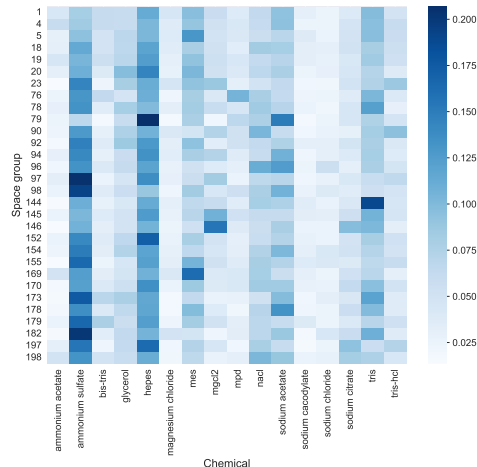


Figure: Correlation of Solvent Content and Compounds. Distributions: Slide: ?? & ??

Relation between Protein Group and Space Group



Relation between Protein Group and Space Group



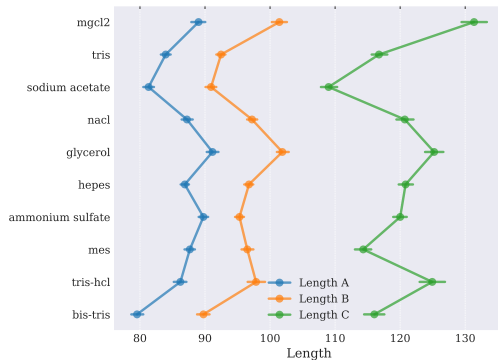
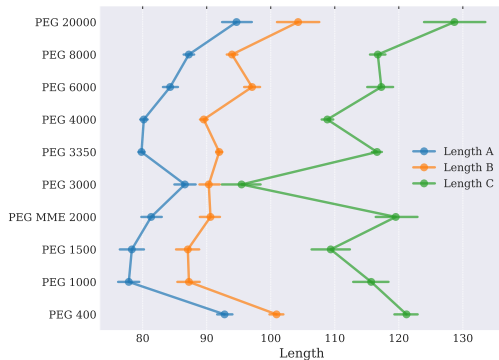


Figure: Relation of Crystal Dimensions and Compounds. Distributions: Slide: ??

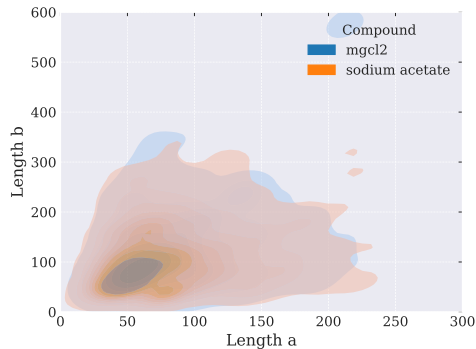
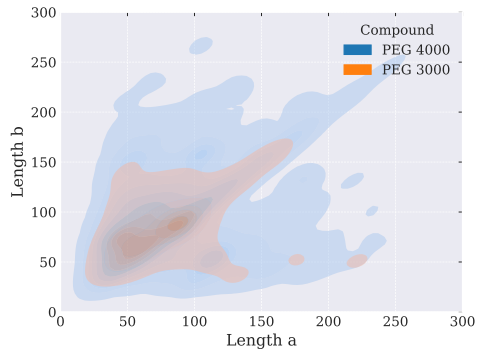
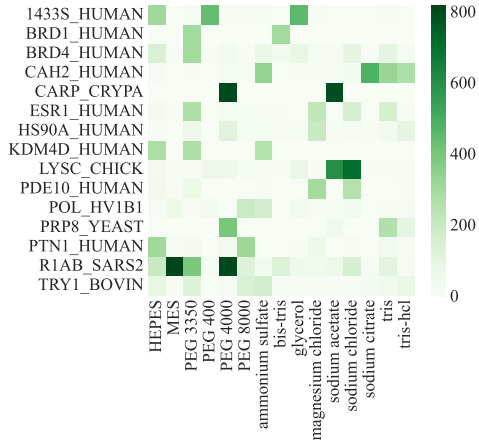
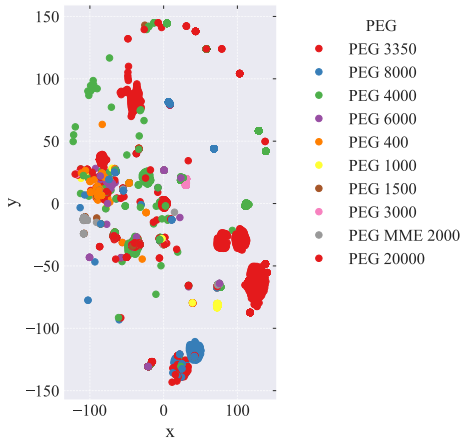


Figure: Relation of Crystal Dimensions and Compounds. Distributions: Slide: ??

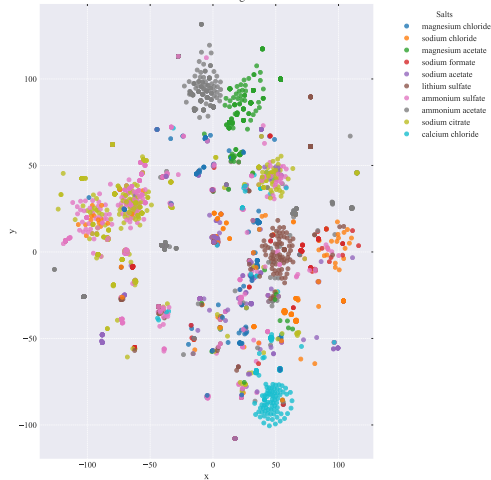
Co-occurrence between Protein and Compound



PEG Embedding



Salts Embedding



- ▶ Beamline Evaluation Platform
- ▶ each plate 96 conditions
- ▶ drops are monitored 1h, 1-3days, 1-4weeks, -4months
- ▶ some images are automatically rated
- ▶ delivers negative data

Plate Setup	Plate Type	Drop Type	Setup Type	User
Wed 01 Oct 2025 10:24	Swissci	Sitting Drop	1 ratio	vpogenberg

Thumbnails

SD30010713 F11-1					
SoFic autoAMP 10 mg/ml					
Shelf	Sample	Conc. mg/ml	Prot. µl	Res. µl	Sec'd µl
1	SoFic autoAMP	10	100	100	0

HH_QIAGEN_JCSG-Core-IV		
Conc.	Component	pH
0.1M	MES	6
30%(v/v)	PEG 600	
5%(w/v)	PEG 1000	
10%(v/v)	Glycerol	

	1	2	3	4	5	6	7	8	9	10	11	12
A												
B												
C												
D												
E												
F												

Inspections					
1 - Vis Full plate true Imager 19 Wed 01 Oct 2025 11:29:51 1 hour after	2 - Vis Full plate true Imager 19 Thu 02 Oct 2025 11:00:50 1 day after	3 - Vis Full plate true Imager 19 Sat 04 Oct 2025 10:49:31 3 days after	4 - Vis Full plate true Imager 19 Wed 08 Oct 2025 11:31:06 1 week after	5 - Vis Full plate true Imager 19 Wed 15 Oct 2025 10:49:04 2 weeks after	6 - Vis Full plate true Imager 19 Wed 29 Oct 2025 10:49:05 4 weeks after

Harvester Manual Harvesting Measure Vis HD



Drop History
Slide Show
Side By Side

Display Images

Vis

Viewing Mode

Quick (Medium Res)
High Res + Active Zoom

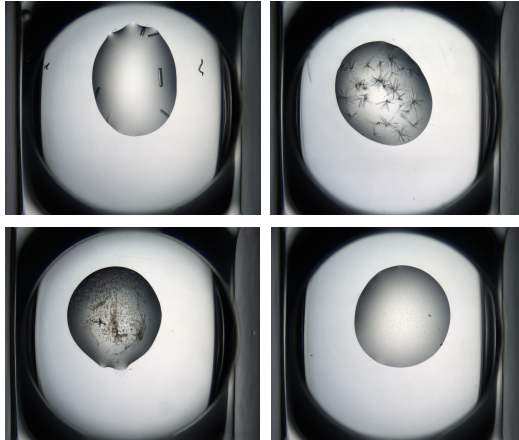
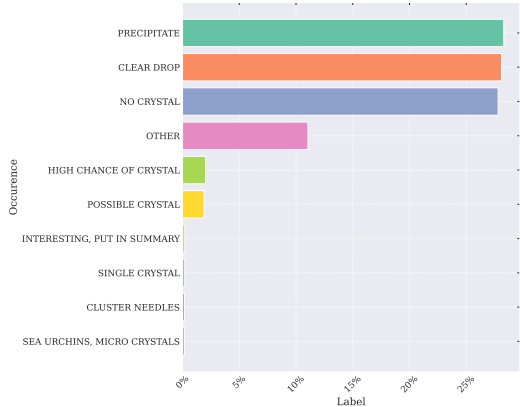
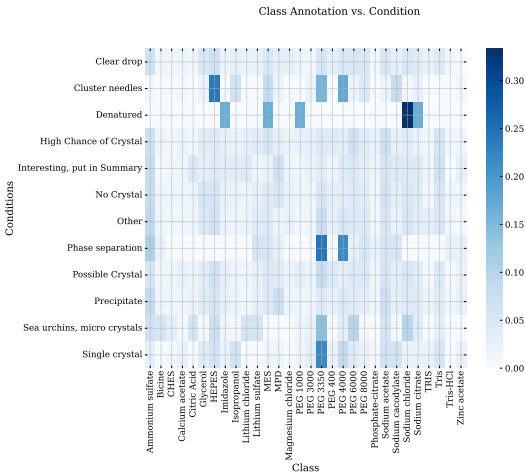
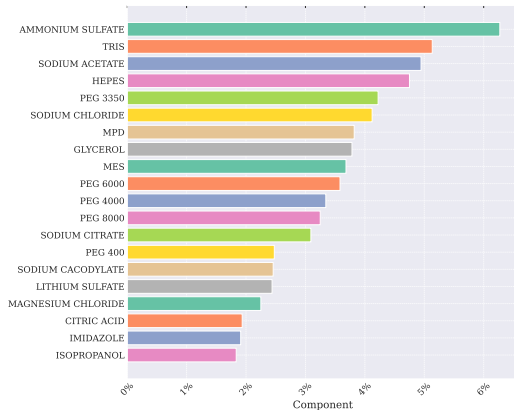
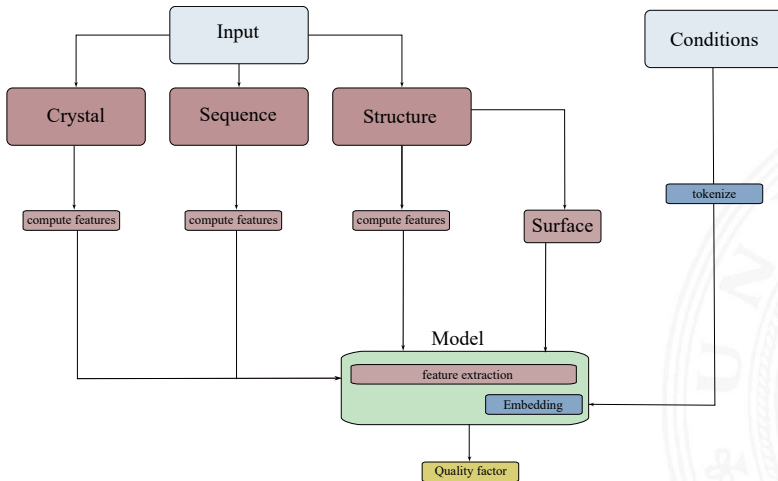


Figure: Scores: 0.99, 0.97, 0.90, 0.3



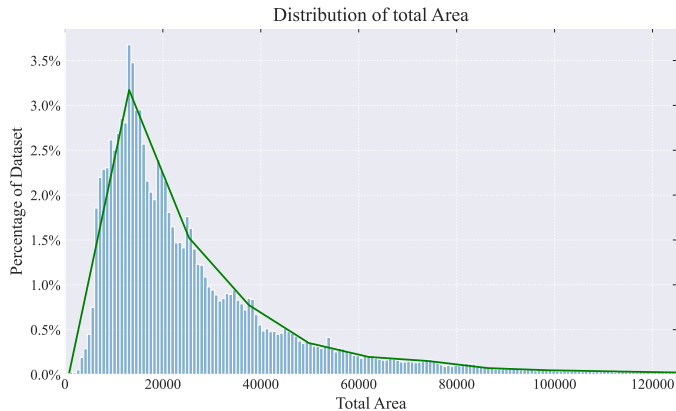
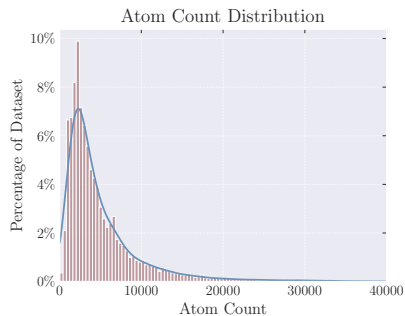


- ▶ while not strongly correlated surface features and embedding seem to have a predictive value of crystallization conditions
- ▶ literature (Liao and Sun, 2025) and EDA indicates that crystal features have also a strong effect on conditions
- ▶ temperature/method space is limited focus more on pH and cocktail composition



- ▶ Pre-train on CRIMS data
- ▶ Task Design
- ▶ Model Design

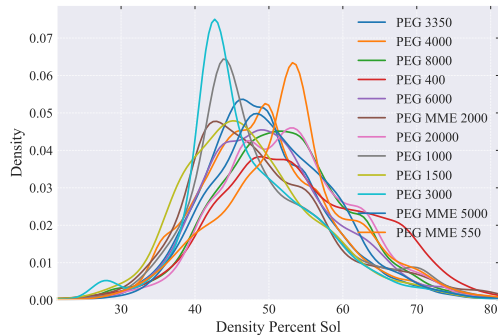




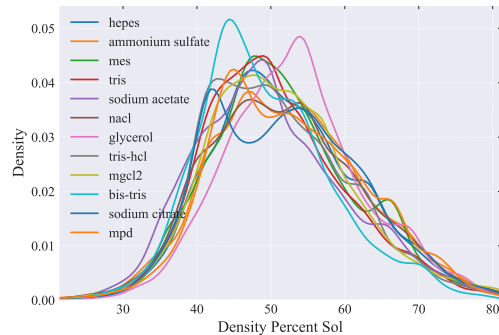
Residue	Codons
Alanine	GCU, GCC, GCA, GCG
Arginine	CGU, CGC, CGA, CGG, AGA, AGG
Asparagine	AAU, AAC
Aspartic acid	GAU, GAC
Cysteine	UGU, UGC
Glutamine	CAA, CAG
Glutamic acid	GAA, GAG
Glycine	GGU, GGC, GGA, GGG
Histidine	CAU, CAC
Isoleucine	AUU, AUC, AUA
Leucine	UUA, UUG, CUU, CUC, CUA, CUG
Lysine	AAA, AAG
Methionine	AUG
Phenylalanine	UUU, UUC
Proline	CCU, CCC, CCA, CCG
Serine	UCU, UCC, UCA, UCG, AGU, AGC
Threonine	ACU, ACC, ACA, ACG
Tryptophan	UGG
Tyrosine	UAU, UAC
Valine	GUU, GUC, GUA, GUG
Stop	UAA, UAG, UGA
Pyrrolysine	UAG
Selenocysteine	UGA
Q or E	CAA, CAG, GAA, GAG
N or D	AAU, AAC, GAU, GAC

base	frequency
G	0.252252
C	0.234234
U	0.234234
A	0.279279

Solvent Content Distribution for different PEGs



Solvent Content Distribution for different Additives

**Figure:** Correlation of Solvent Content and Compounds.

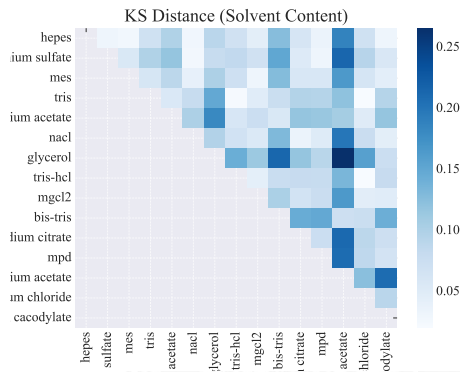
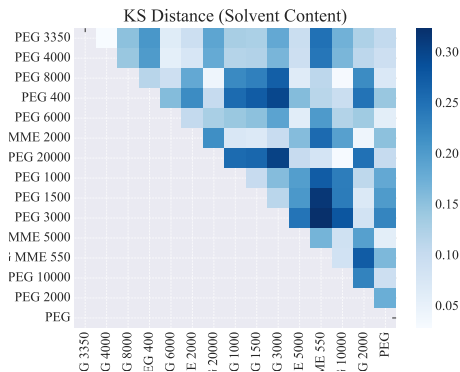


Figure: KS Distance between Compounds

